



北京航空航天大学
BEIHANG UNIVERSITY



飞行学院
School of Aviation

飞行原理 Principles of Flight

第二章 低速空气动力学

2-4 三维机翼与全机低速空气动力

2025年9月28日星期日

目录

1 机翼三维效应及诱导阻力

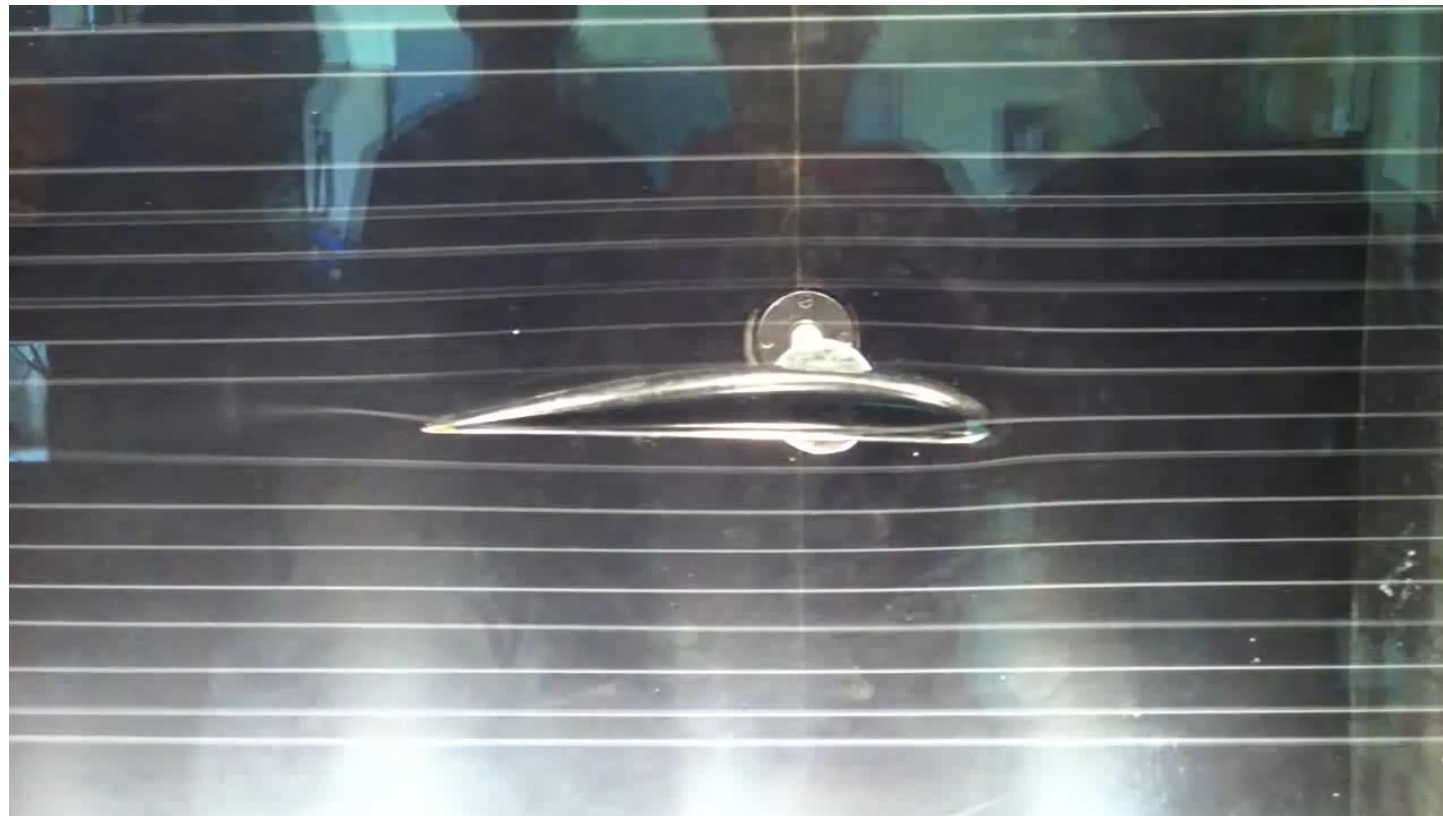
2 机翼几何参数及其对气动影响

3 全机低速气动特性

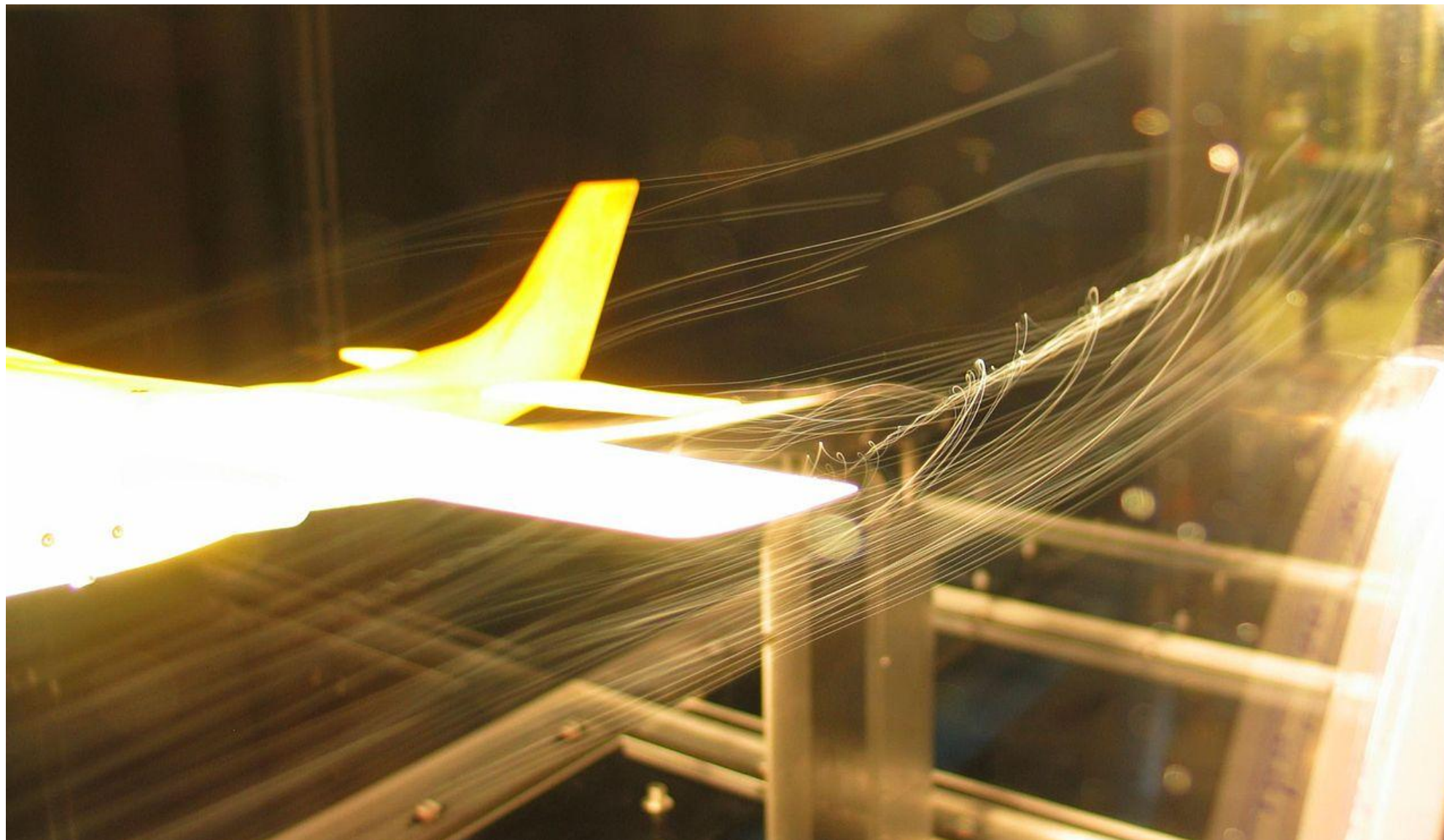
大展弦比直机翼低速扰流



<https://youtu.be/WxXtVLjDKCo>



大展弦比直机翼低速扰流



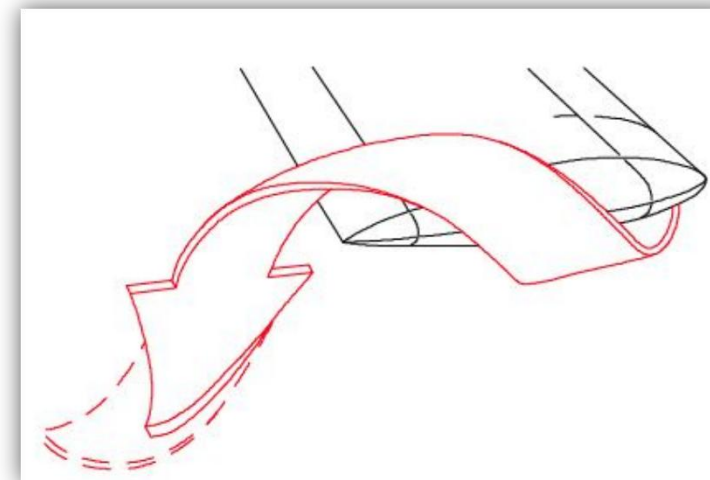
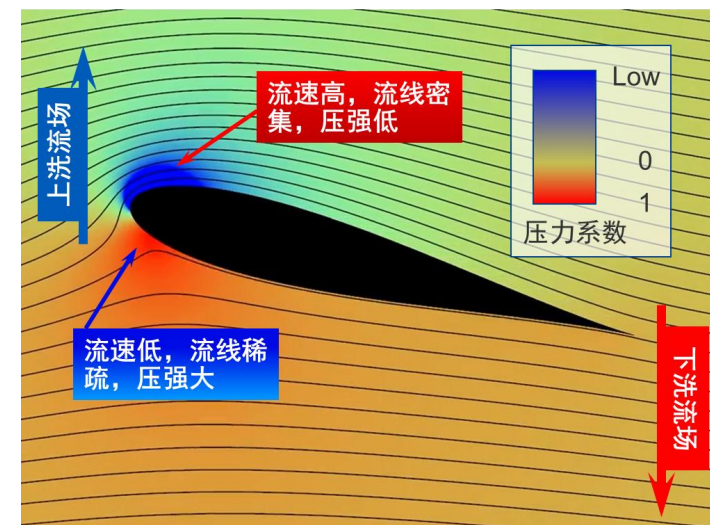


<https://youtu.be/c-Eab3bukZM>

翼尖涡的形成



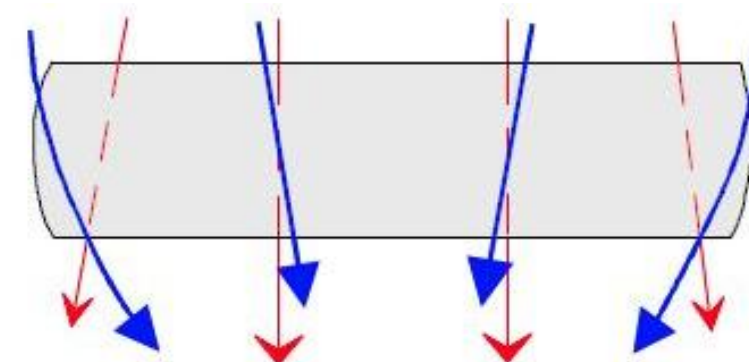
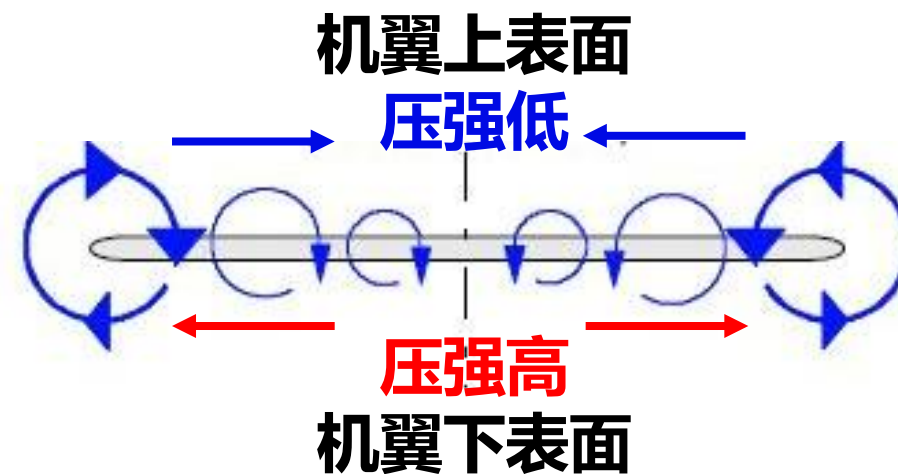
- 正迎角下机翼上、下表面压差形成升力
- 有限翼展机翼翼尖部分形成从下表面高压区从侧面翻向上表面低压区的**翼尖涡(Wingtip Vortex)**
- **翼尖涡是造成二维翼型与三维机翼气动差异的根本原因**
- 二维翼型可视为无限翼展的机翼，由于无翼端存在，上下翼面的压差不会引起展向的流动，展向任一剖面均保持二维翼型的特性



尾涡和尾涡面的生成



- 翼尖涡带动下，上翼面的流线向对称面偏斜，下翼面流线向翼尖方向倾斜
- 上下翼面展向流动带动机翼后缘形成与翼尖涡方向一致的**尾涡/后涡缘 (Trailing vortex)**
- 单独尾涡弱于翼尖涡，全部尾涡共同形成尾涡面 (Trailing vortex sheet)
- 随迎角增大翼尖涡和尾涡共同增强

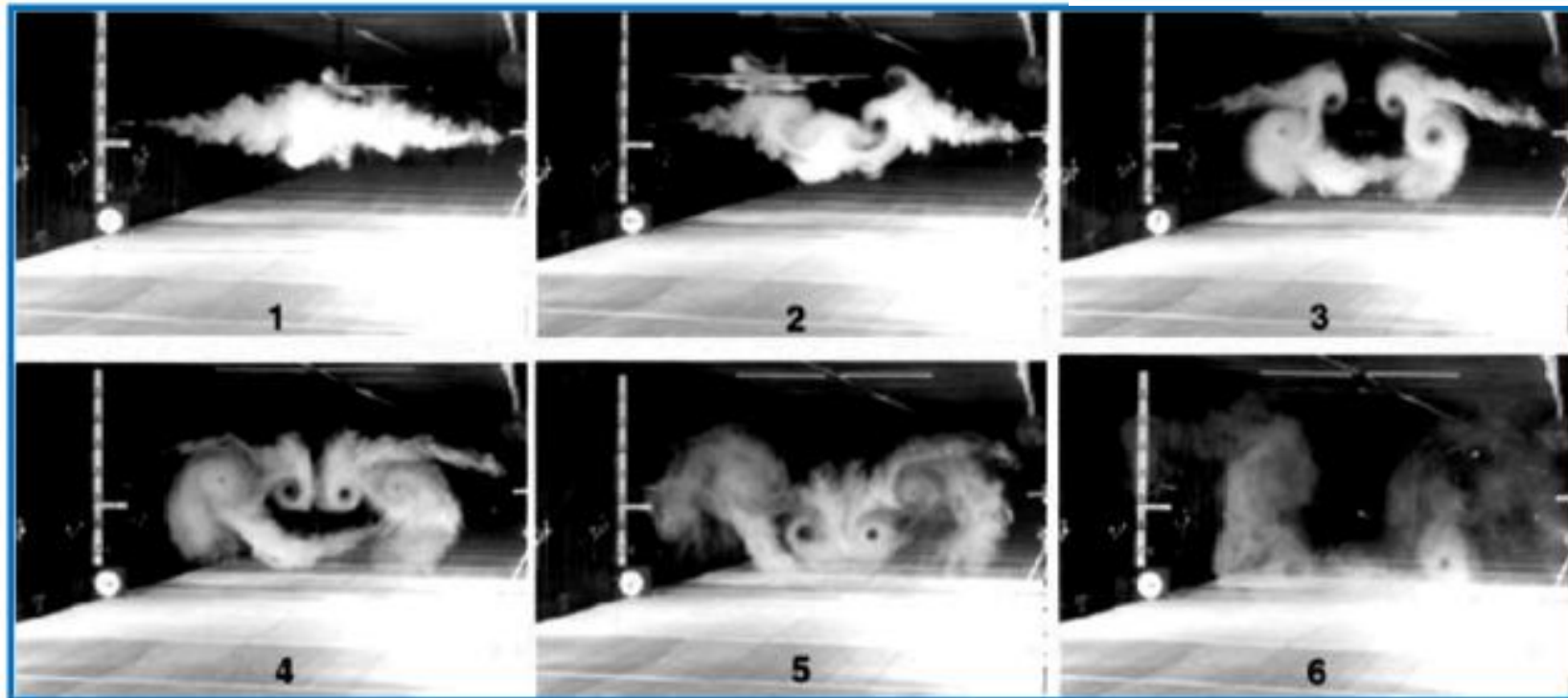


蓝色：上翼面流线
红色：下翼面流线

尾涡和尾涡面的生成



Boeing747模型的尾涡实验 (in Langley 14- by 22-Foot Tunnel)

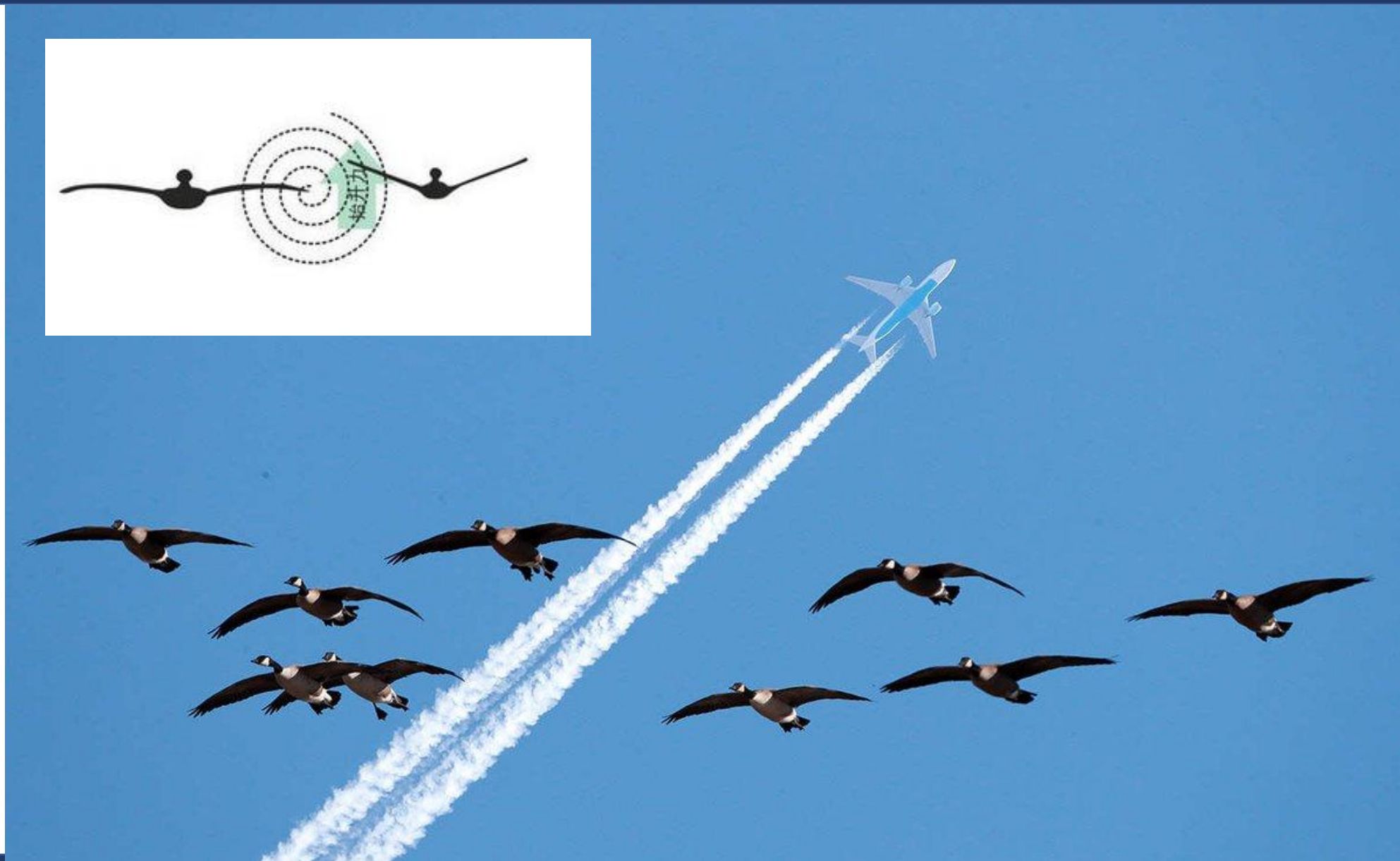
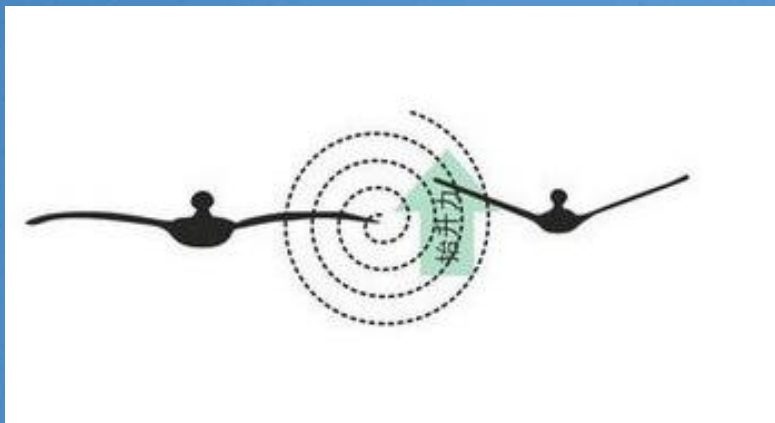


尾涡和尾涡面的生成



Boeing747在降落过程中，其尾涡对烟囱流场的影响

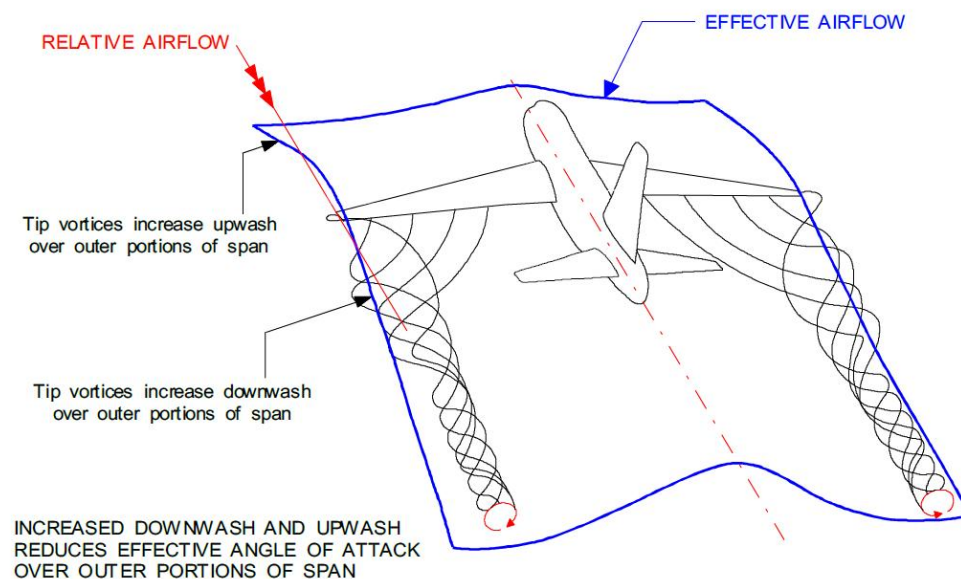
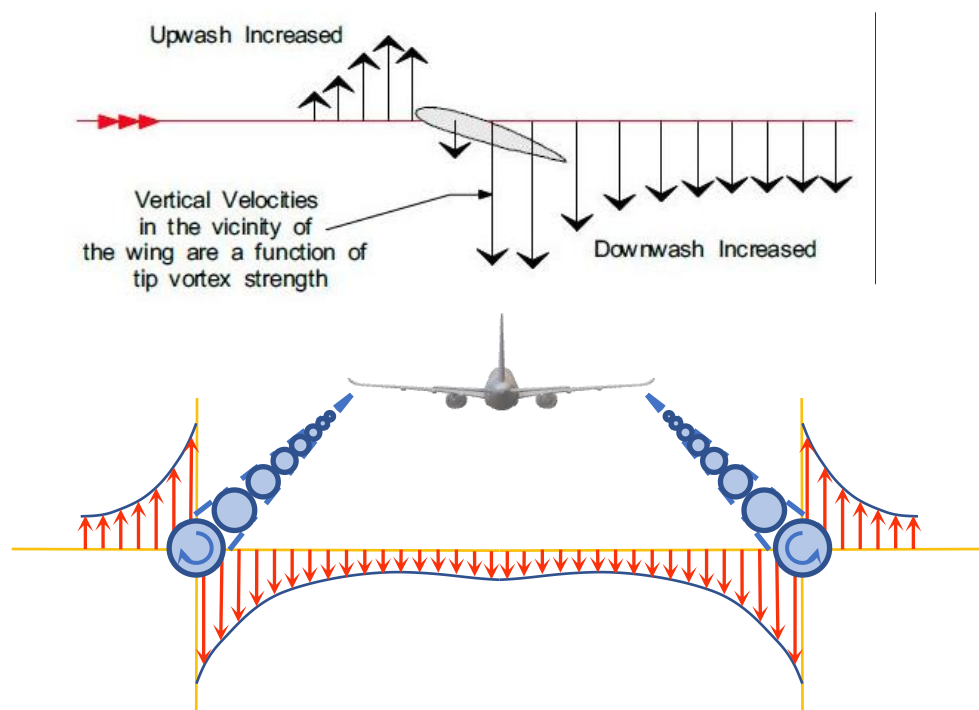
尾涡和尾涡面的生成



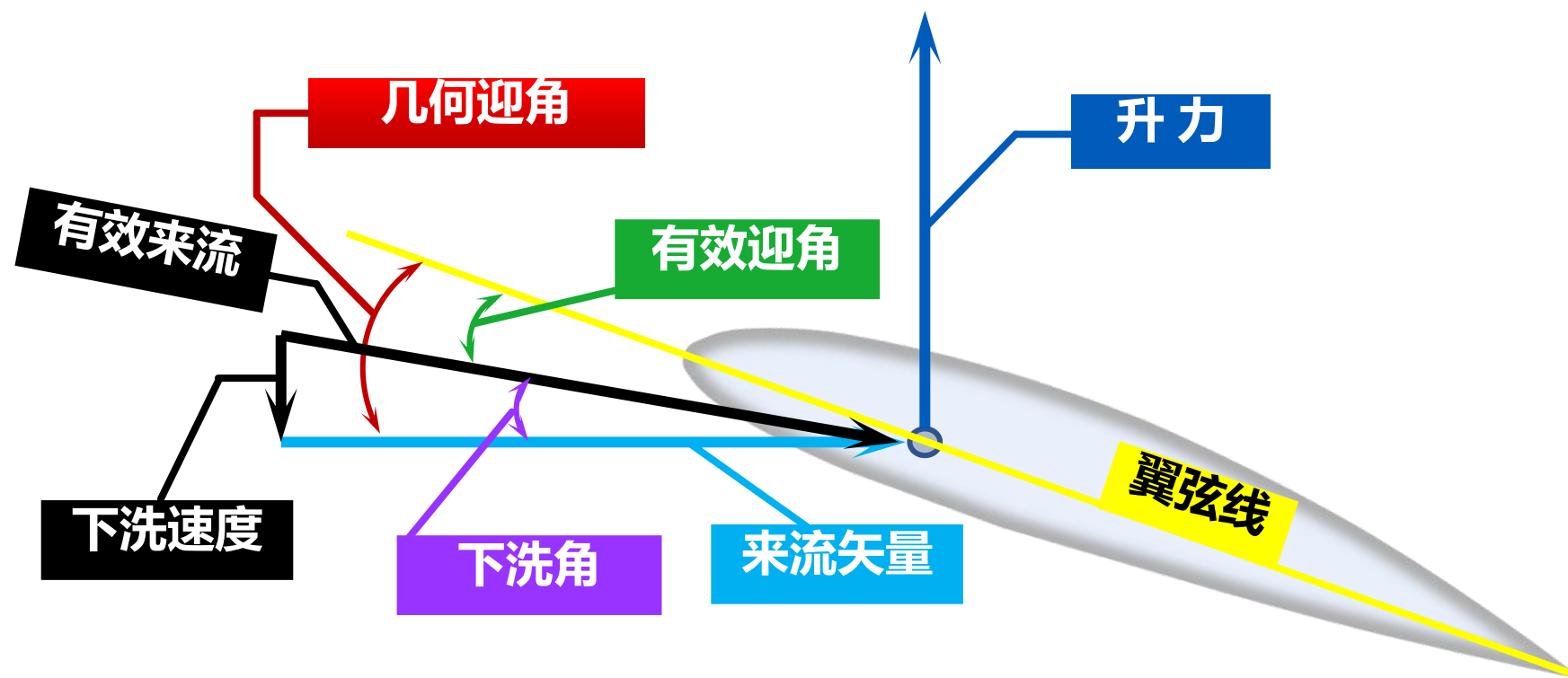
翼尖涡的诱导流场



- 机翼二维翼型截面产生升力时前方形形成上洗流场，后方形形成下洗流场
- 翼尖涡和后缘涡造成涡内侧产生下洗流动，外侧产生上洗流动
- 随迎角增加二维翼型和翼尖/后缘涡引起的诱导速度均会增大

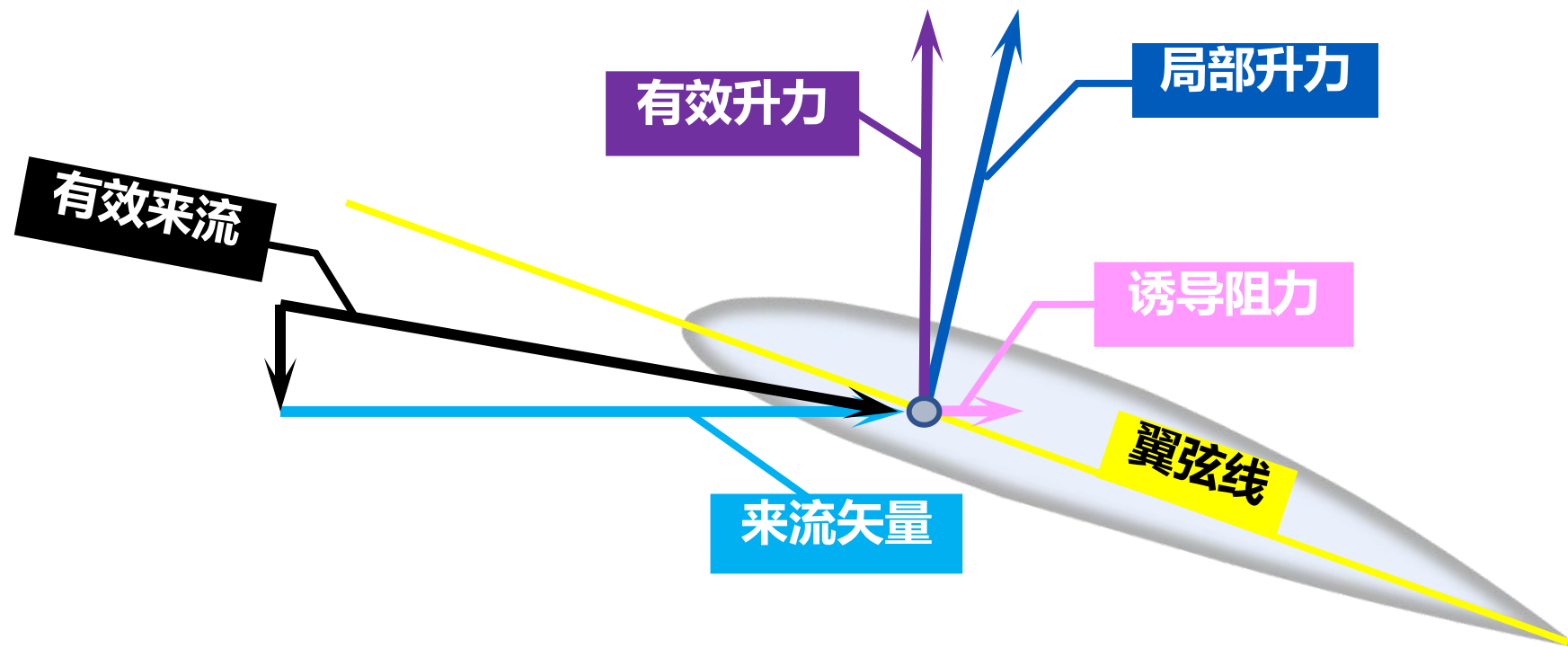


下洗流场对气动力的影响



- 下洗速度与原始来流速度做矢量和得到有效来流
- 有效来流与原始来流矢量夹角为**下洗角** (Induced Angle of Attack)
- 有效来流与翼弦线夹角为**有效迎角** (Effective Angle of Attack)

下洗流场对气动力的影响



- 下洗速度造成有效迎角减小，有效来流方向较自由来流矢量向下偏折
- 局部升力垂直于有效来流方向，将其分别投影到原始来流的平行方向分量即为**诱导阻力 (Induced Drag)**
- 升力方向偏折和有效迎角降低，造成三维机翼升力线斜率低于二维翼型

Induced drag is the result of (诱导阻力是由于:)

- A** Downwash generated by tip vortices
(翼尖涡产生的下洗流)
- B** Propeller slipstream over the wing
(流过机翼的螺旋桨滑流)
- C** Boundary layer separation on the aft portion of the wing
(机翼后缘的边界层分离)
- D** Aerodynamic interaction between aeroplane parts
(机翼各部件之间的相互气动干扰)

提交

Induced drag is the result of

- ☒ A Downwash generated by tip vortices
- ☐ B Propeller slipstream over the wing
- ☐ C Boundary layer separation on the aft portion of the wing
- ☐ D Aerodynamic interaction between aeroplane parts

提交

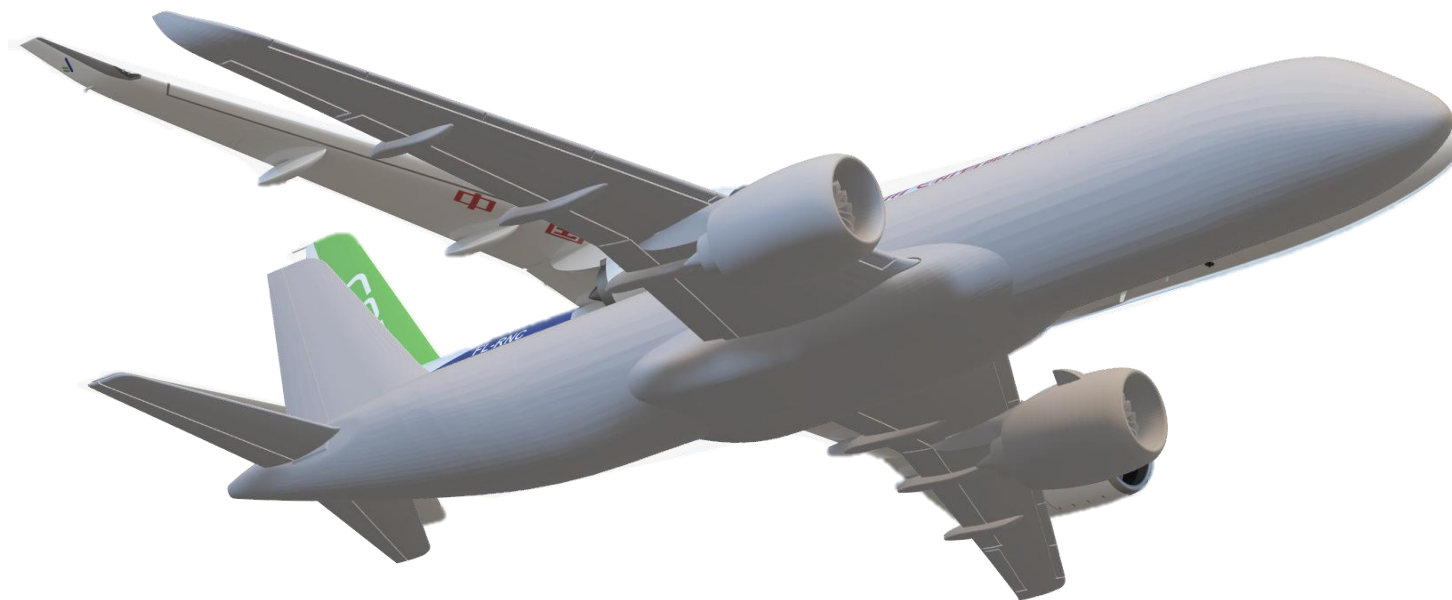
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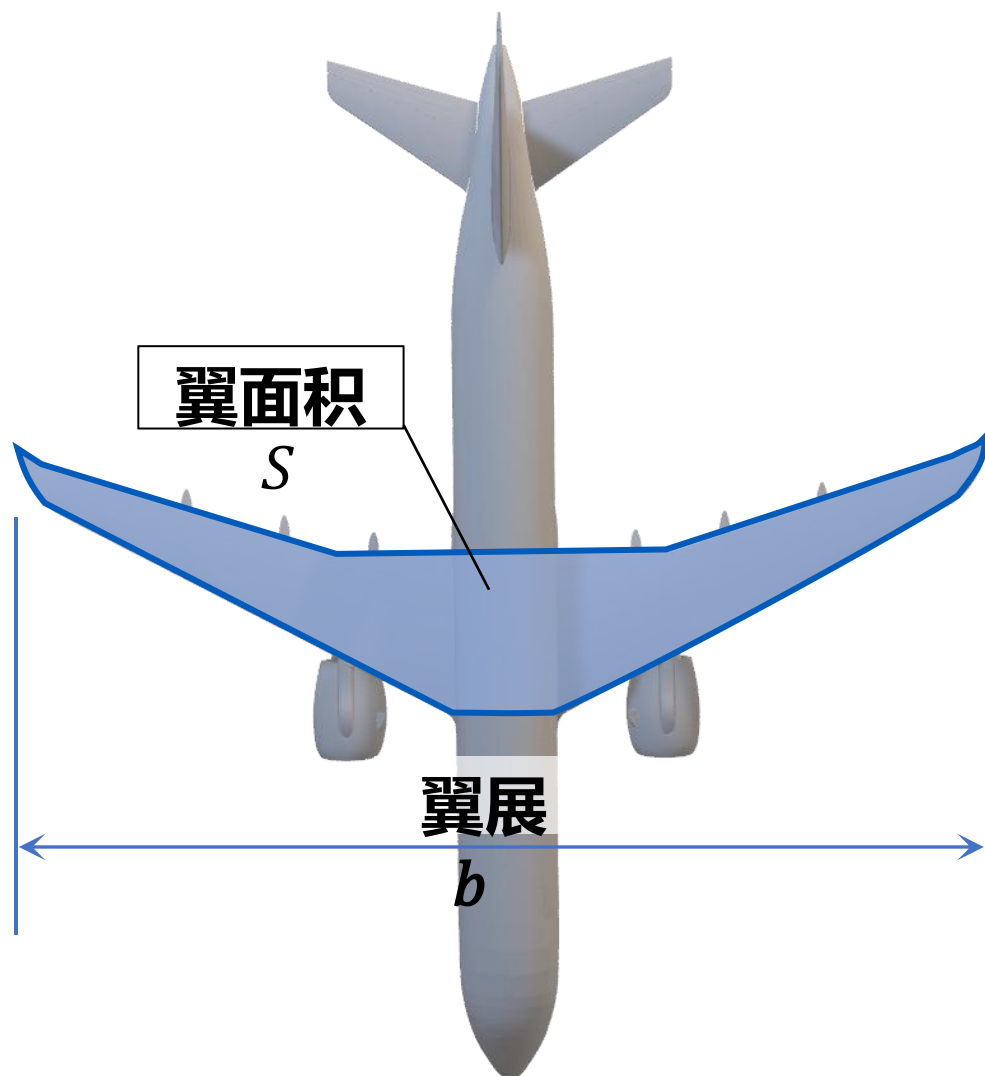
3 全机低速气动特性

机翼的几何参数





机翼几何参数的气动影响



- 翼面积 (Wing Area) S

→ 机翼在水平面内的投影面积

- 翼展 (Span) b

→ 机翼左右翼尖之间的长度

- 展弦比 (Aspect ratio):

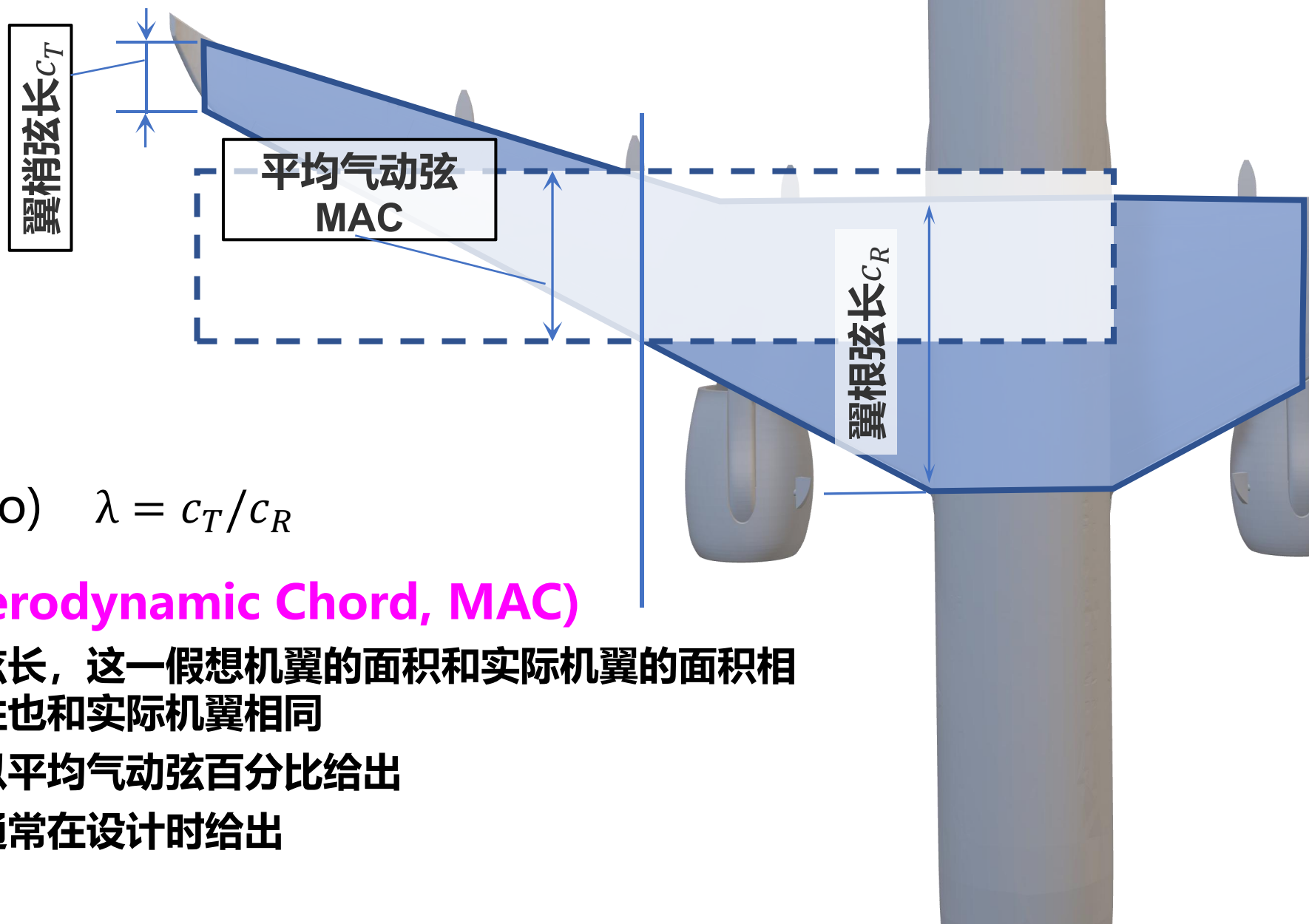
→ 翼展平方除以面积, 描述机翼的细长程度 $A_R = \frac{b^2}{S}$

战斗机	大型客机	滑翔机
2~3	12~20	~35

- 翼载荷 (Wing loading)

→ 飞机质量和机翼面积之 W/S

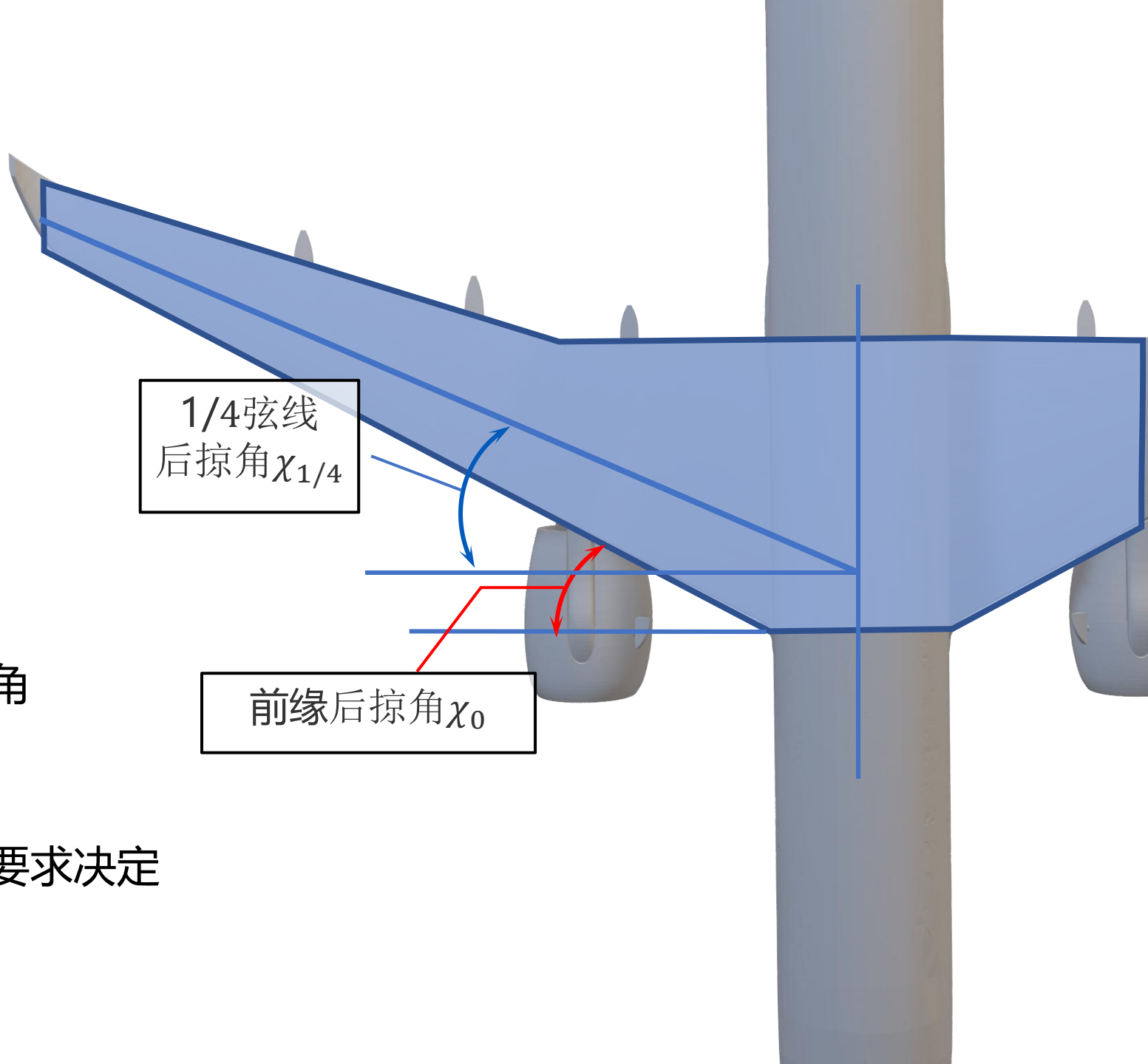
PA-28	70 kg/m ²
Spitfire	135 kg/m ²
Learjet 31	286 kg/m ²
B737-900	675 kg/m ²

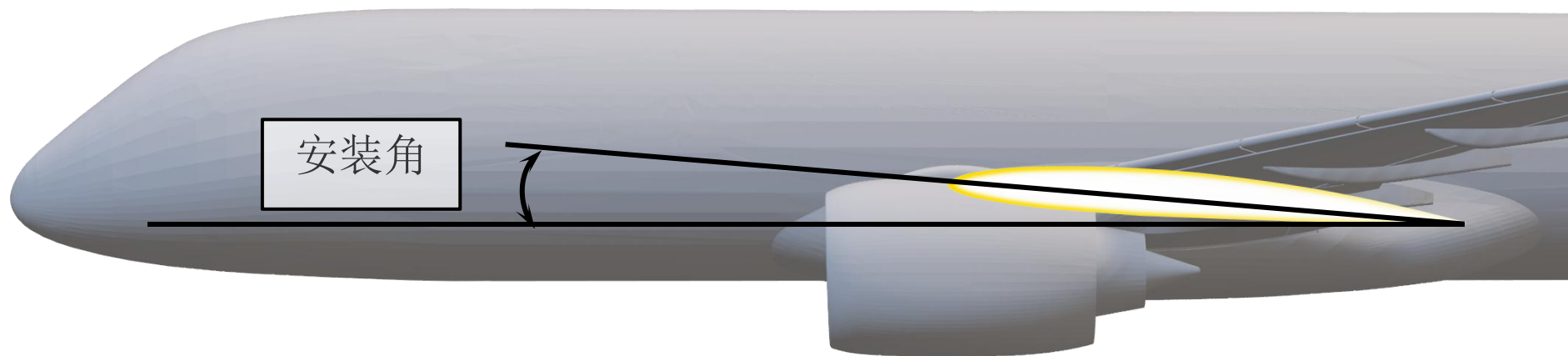


- 弦长 c
 - 翼梢弦长 c_T
 - 翼根弦长 c_R
- 尖削比/梯形比(Taper ratio) $\lambda = c_T/c_R$
- 平均气动弦(Mean Aerodynamic Chord, MAC)
 - 等效假想矩形机翼的弦长，这一假想机翼的面积和实际机翼的面积相同，其升力和力矩特性也和实际机翼相同
 - 飞机的重心位置都会以平均气动弦百分比给出
 - 平均气动弦所在位置通常在设计时给出

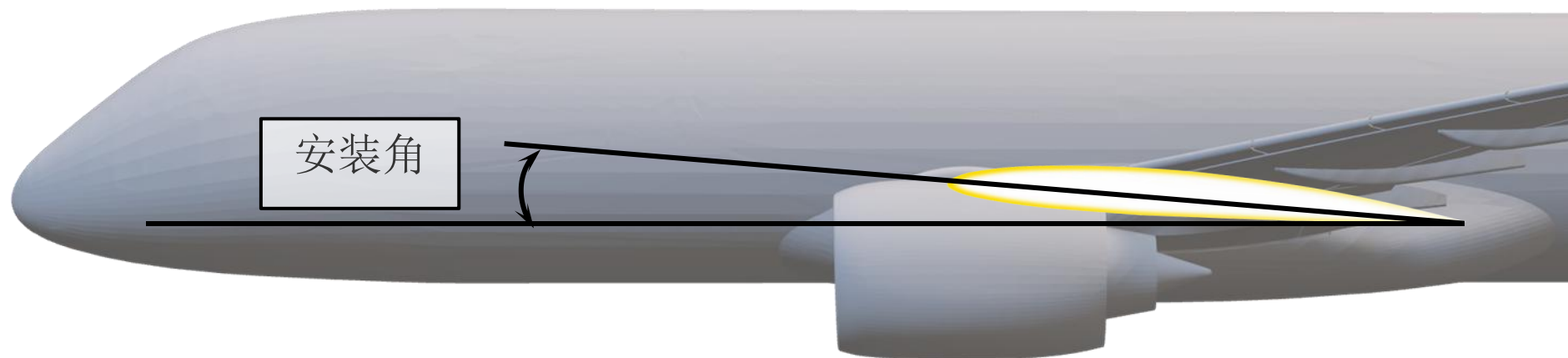
- 后掠角(Sweep angle)

- ➔ 机翼与机身轴线的垂线之间的夹角
- ➔ 前缘后掠角 χ_0 (影响高速性能)
- ➔ $1/4$ 弦线弦长 $\chi_{1/4}$ (影响低速性能)
- ➔ 后掠角主要由飞机的跨声速性能要求决定
- ➔ 同时影响飞机的升力和失速特性

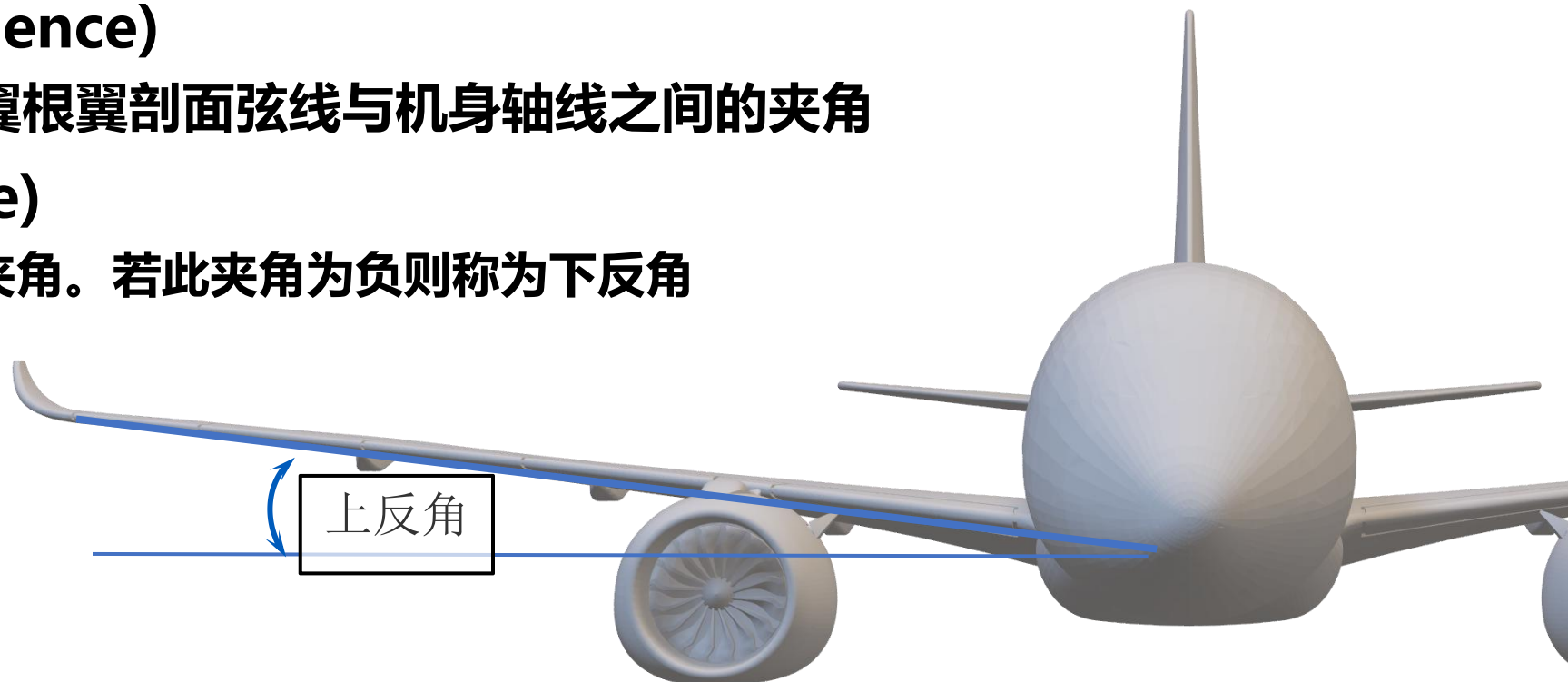




- 安装角(Angle of incidence)
 - 机翼安装在机身上时翼根翼剖面弦线与机身轴线之间的夹角



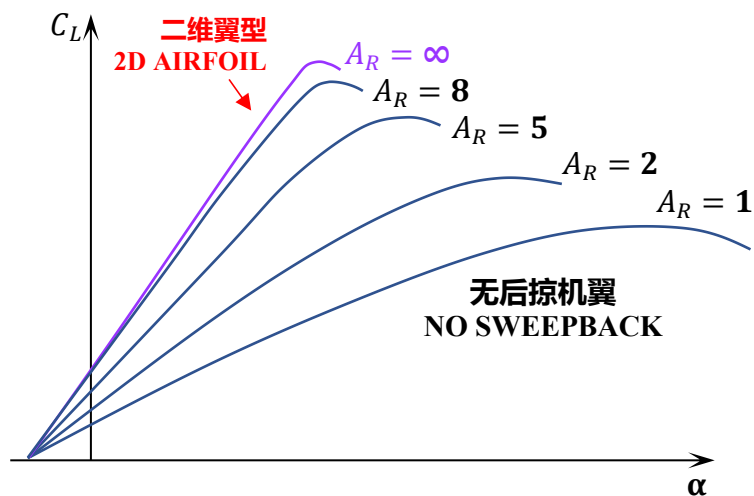
- **安装角(Angle of incidence)**
 - 机翼安装在机身上时翼根翼剖面弦线与机身轴线之间的夹角
- **上反角(Dihedral angle)**
 - 机翼基准面和水平面的夹角。若此夹角为负则称为下反角



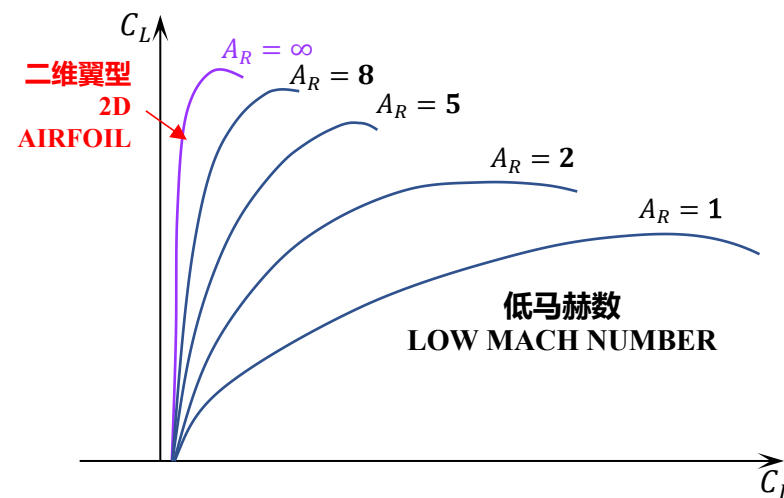


展弦比对机翼气动特性影响

- 展弦比增加将显著减少翼尖涡影响的区域，增加机翼的升力线斜率和最大升力线斜率，降低诱导阻力
- 展弦比增加的缺点
 - 增加机翼根部弯矩
 - 降低飞机滚转机动性
 - 容易造成起降过程翼尖蹭地



A_R —展弦比



大展弦比机翼带来的问题



美国U-2
侦察机

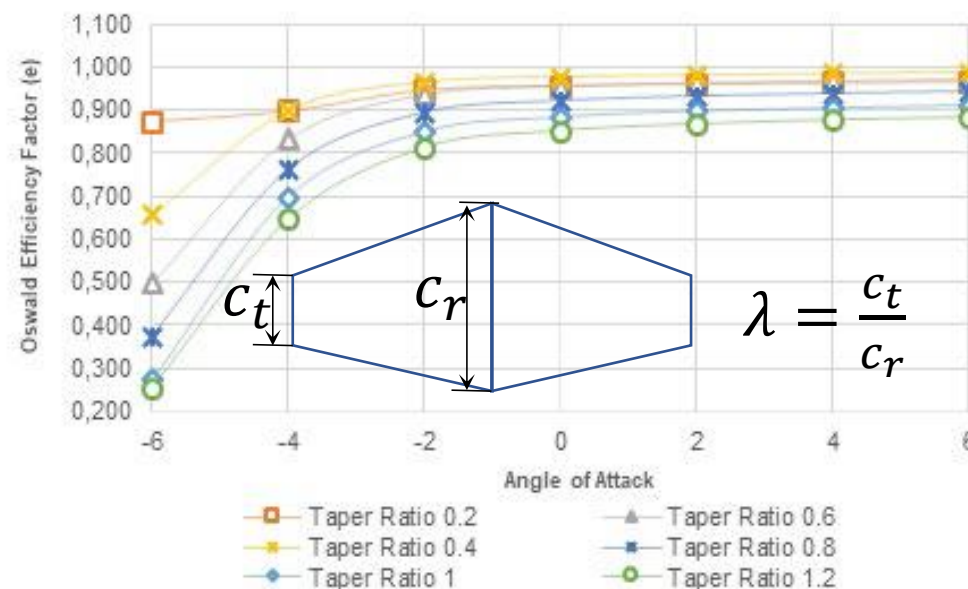
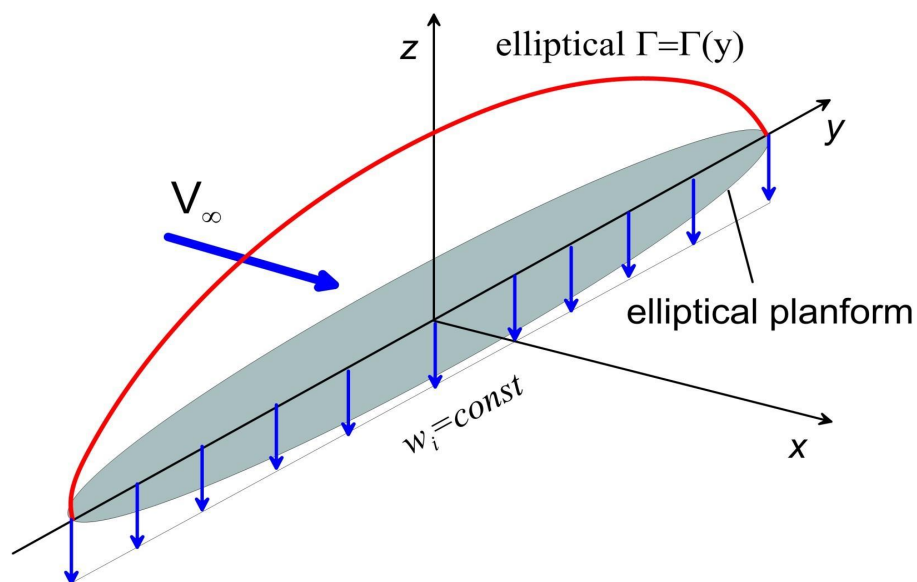
降落时翼
尖蹭地



平面形状对机翼气动影响



- 根据普朗特升力线理论，**升力延翼展方向呈椭圆分布时诱导阻力最小**
 - 仅椭圆平面形状机翼在任何迎角下满足椭圆升力分布
 - 调整梯形机翼的尖削比 λ 可使其在一定迎角范围内获得近似椭圆升力分布
 - 展向升力分布与椭圆分布的近似程度可用奥斯瓦尔德系数 e_0 (Oswald efficiency factor)衡量



经典的椭圆机翼——英国喷火战斗机







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Welcome to the Spitfire AA810 project page, a project dedicated to the memory of all the men of the Photographic Reconnaissance Unit who risked, and gave, their lives during the Second World War.

Flying deep into enemy territory, mostly unarmed and devoid of protection of any sort, these men photographed the movements of enemy supplies, men, ships and planes to gain the vital intelligence for Allied Commanders to plot the strategy of the war.

The Photographic Reconnaissance Unit (PRU) worked for all the branches of the armed forces in all theatres bringing home the crucial up to date information required to keep the Allies ahead.

Largely the PRU's efforts have disappeared into history and with the levels of automation in the development of satellite imaging and unmanned aerial vehicle technology we are unlikely to ever send crews to carry out these missions again.

Few of the men involved have been credited with the work they carried out, this small band of brothers, who suffered considerable casualties and, due to the nature of their missions, those that were lost will most likely never be found. Now, fortunately, one of those lost Spitfires has been found and is under rebuild to flying condition, so the stories of the men that flew it can be told. Through this venture we hope to raise more awareness worldwide of the sacrifices the PRU made.



100 Years of Women in Law .
Manchester Law Society . GIN



90 Years Of The Cambridge
Aero Club



Nottinghamshire Law Society .
100 Years of Women in Law .
GIN



Spitfire Heritage Pin



Glassware (UK only)

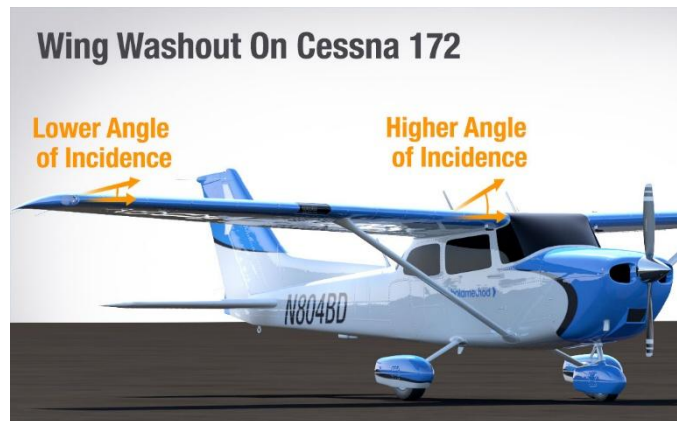


Spitfire AA810 Benefactor
Certificate

其他诱导阻力降低方法



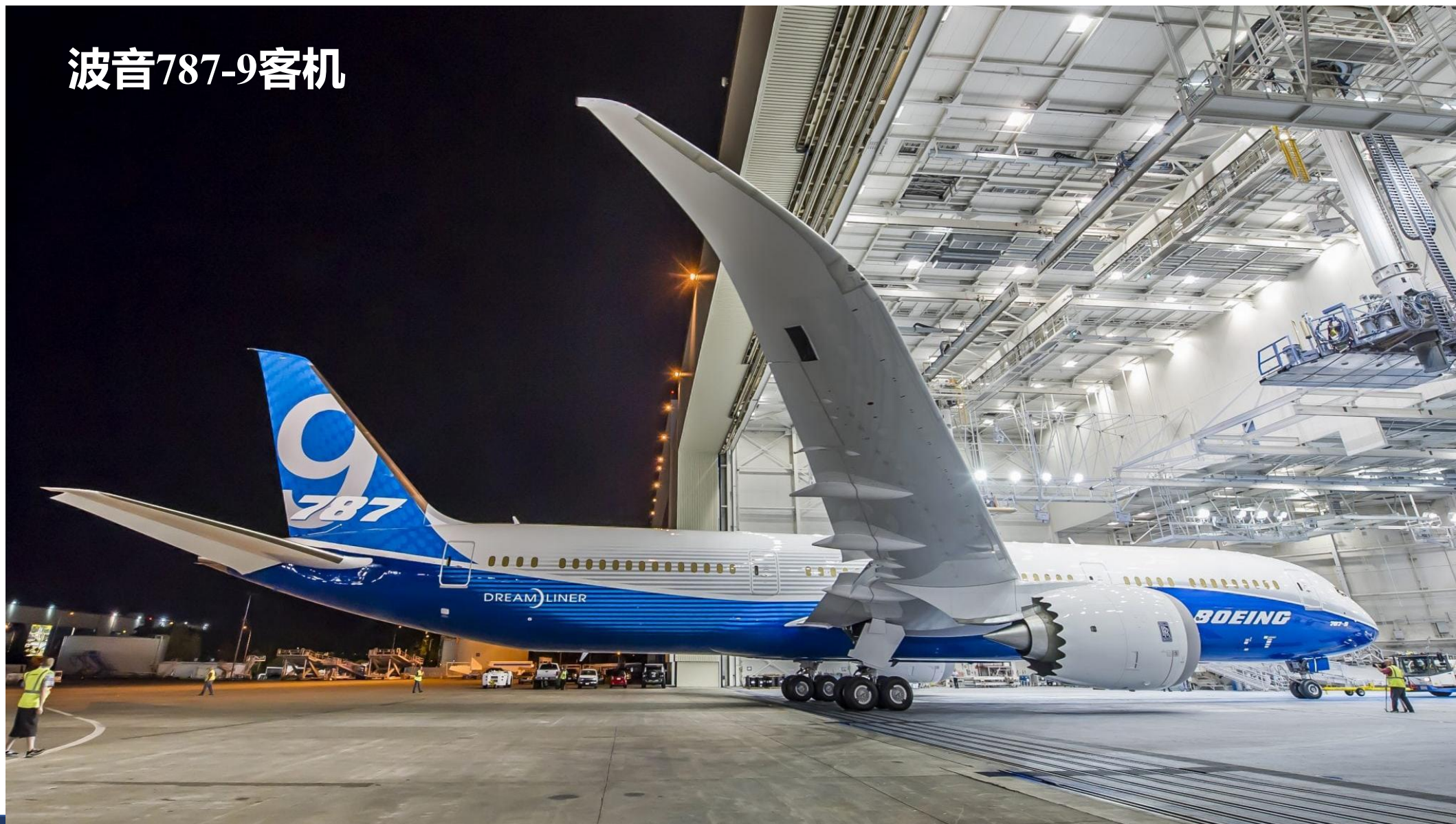
- **机翼扭转**：通过沿展向改变翼型迎角（外洗/Washout）或弯度的方式调整升力展向分布
- **翼梢端板**：放置在机翼尖端的平板以减小翼尖涡，从而一定程度上降低诱导阻力
- **翼尖油箱**：除减小翼尖涡外为机翼弯曲应力卸载
- **翼梢小翼**：“阻挡”来自下翼面的气流以减小翼尖涡
- **翼尖修型**：修改翼尖几何形状抑制翼尖涡产生



787客机的斜削式翼梢小翼



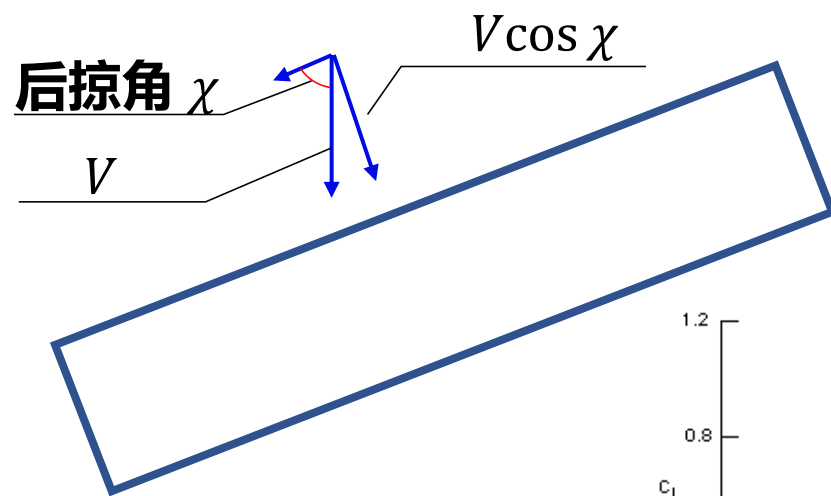
波音787-9客机



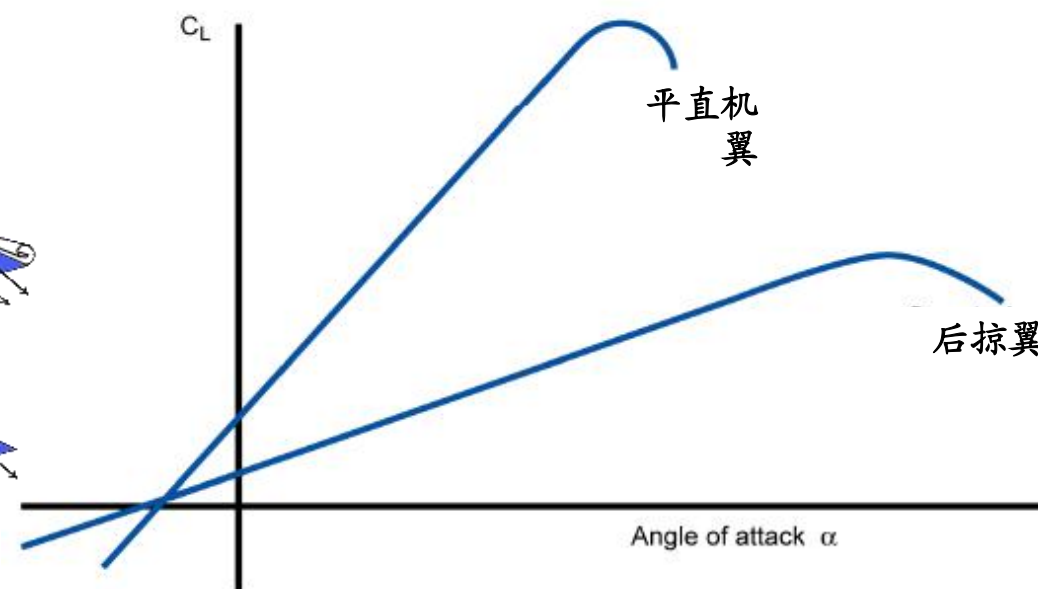
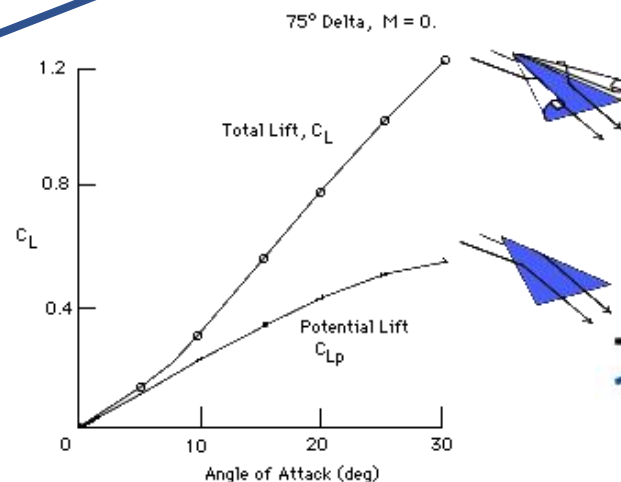


后掠角对机翼升阻特性影响

- 降低有效来流速度降低马赫数增加带来的阻力影响
 - 诱导阻力与同翼展、同展向升力分布平直翼相同*
 - 中小迎角下降低升力线斜率
 - 平直机翼的最大升力系数更大，升力系数曲线斜率越大，临界迎角越小



*Munk's Stagger Theorem



Modifying the shape of which part of the aeroplane would have the largest effect on induced drag? (改变飞机哪部分的形状对诱导阻力影响最大?)

- ☐ A Tailplane (尾翼)
- ☐ B Wing root junction (翼根连接处)
- ☐ C Flaps (襟翼)
- ☐ D Wing tip (翼尖)

提交

Modifying the shape of which part of the aeroplane would have the largest effect on induced drag?

- ☐ A Tailplane
- ☐ B Wing root junction
- ☐ C Flaps
- ☒ D Wing tip

提交

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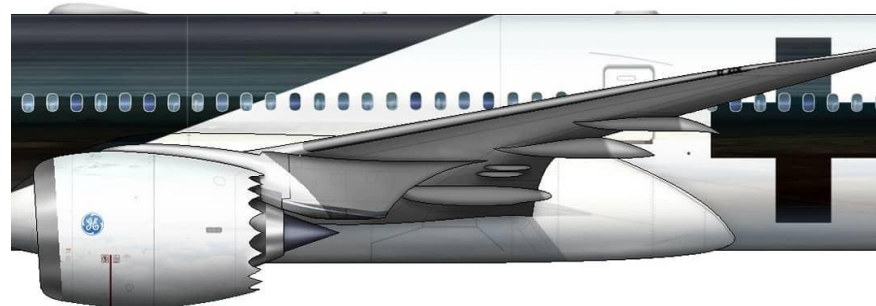
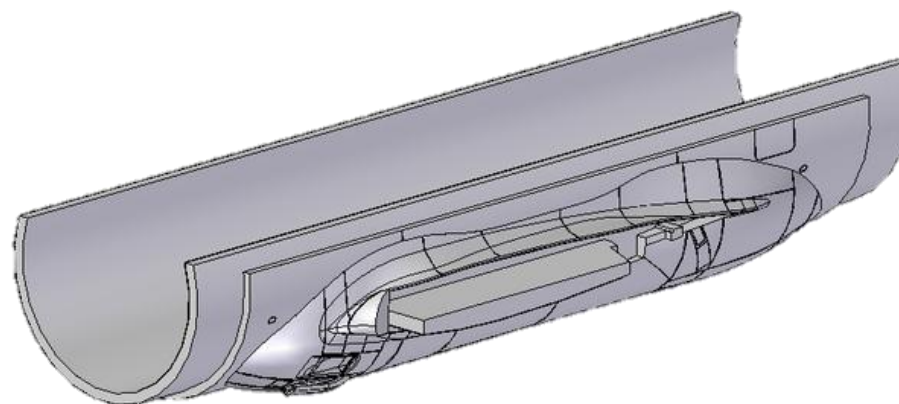
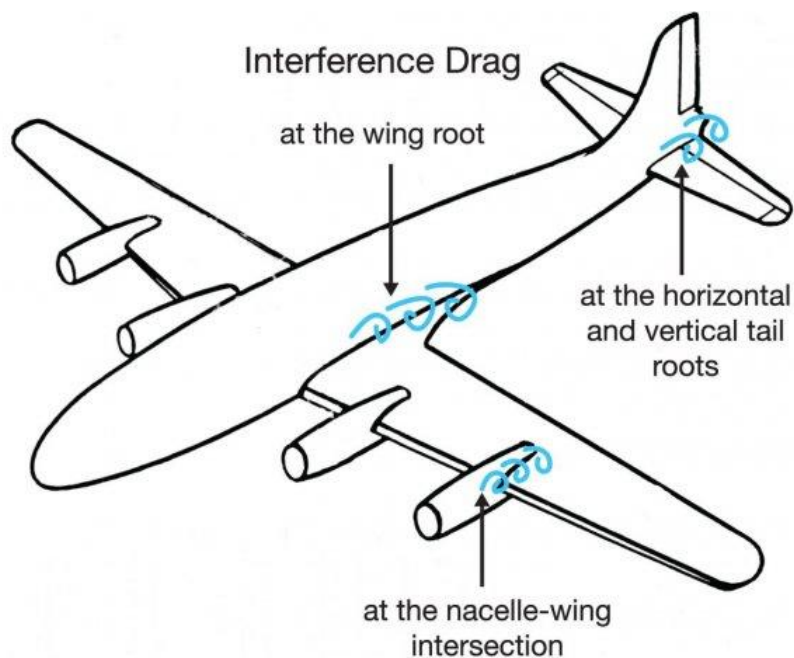
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3 全机低速气动特性

干扰阻力的产生与抑制



- 干扰阻力(Interference Drag)是飞机各结构不规则连接处的边界层相互干扰产生的额外阻力
- 主要依靠气动修型和整流罩解决



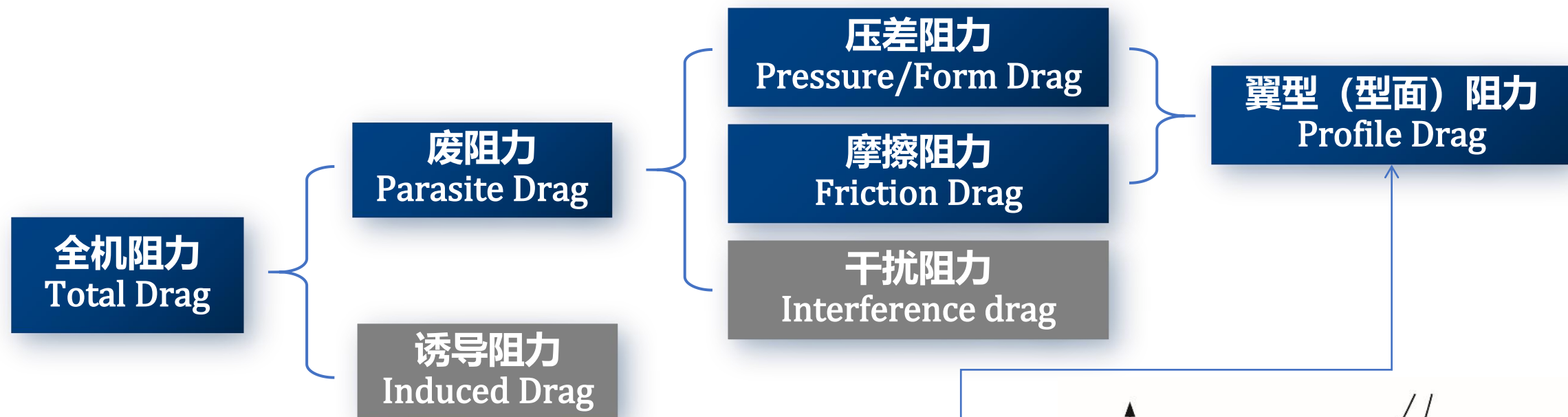
翼身融合设计的性能影响



- 通过将翼身融合设计为整体最大限度减少翼身之间形成干扰阻力并减少浸润面积而降低摩擦阻力
- 增大飞机前向投影面积，有可能带来压差阻力和跨/超声速阻力的增加

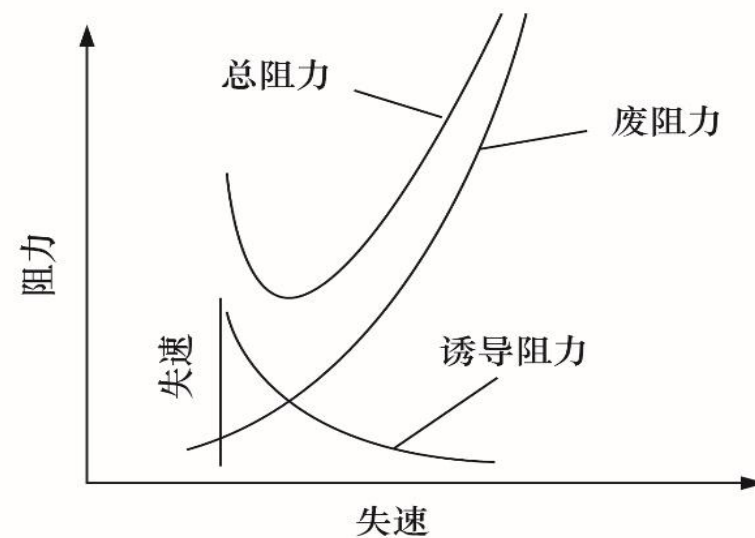


全机的阻力构成



- 平飞时，重力=升力
- 废阻力与动压（速度平方）呈正比
- 诱导阻力与动压（速度平方）呈反比

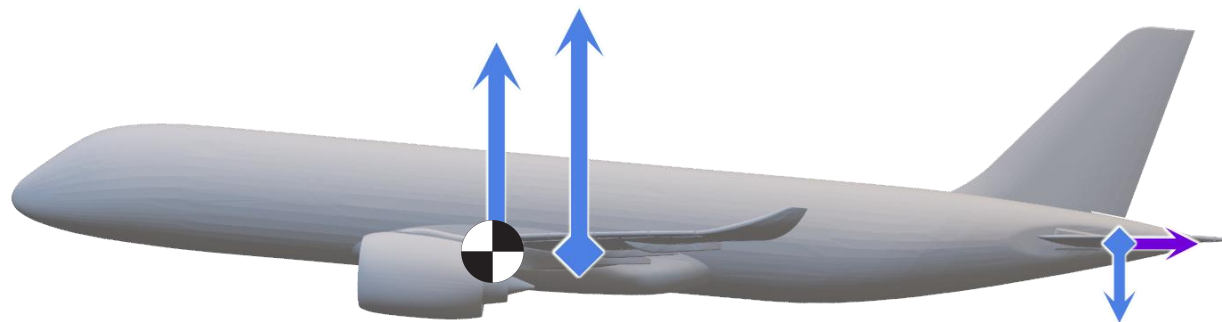
$$D = \frac{kW^2}{q_\infty S} - q_\infty C_{D\Diamond} S$$



平尾对全机升阻特性的影响



- 通过调整平尾升力实现不同迎角下的俯仰力矩平衡即俯仰力矩配平 (Trim)
 - 尾翼升力对全机总升力产生影响
 - 尾翼产生升力时造成的诱导阻力即**配平阻力**



飞机其他部件气动力影响



- 机身/尾翼/吊挂等部件、机载系统的进排气通道，飞机的构型变化（收放起落架、襟翼等）等均会产生一定的废阻影响
- 采用翼身融合设计的机身可提供一定升力





- **升阻比 (Lift Drag Ratio)** 是衡量飞机气动效率的最直接指标, 定义为同一迎角下升力与阻力的比值

$$K = \frac{L}{D} = \frac{C_L}{C_D}$$

- 升阻比越大意味着同样重量下维持平飞所需的推力越小

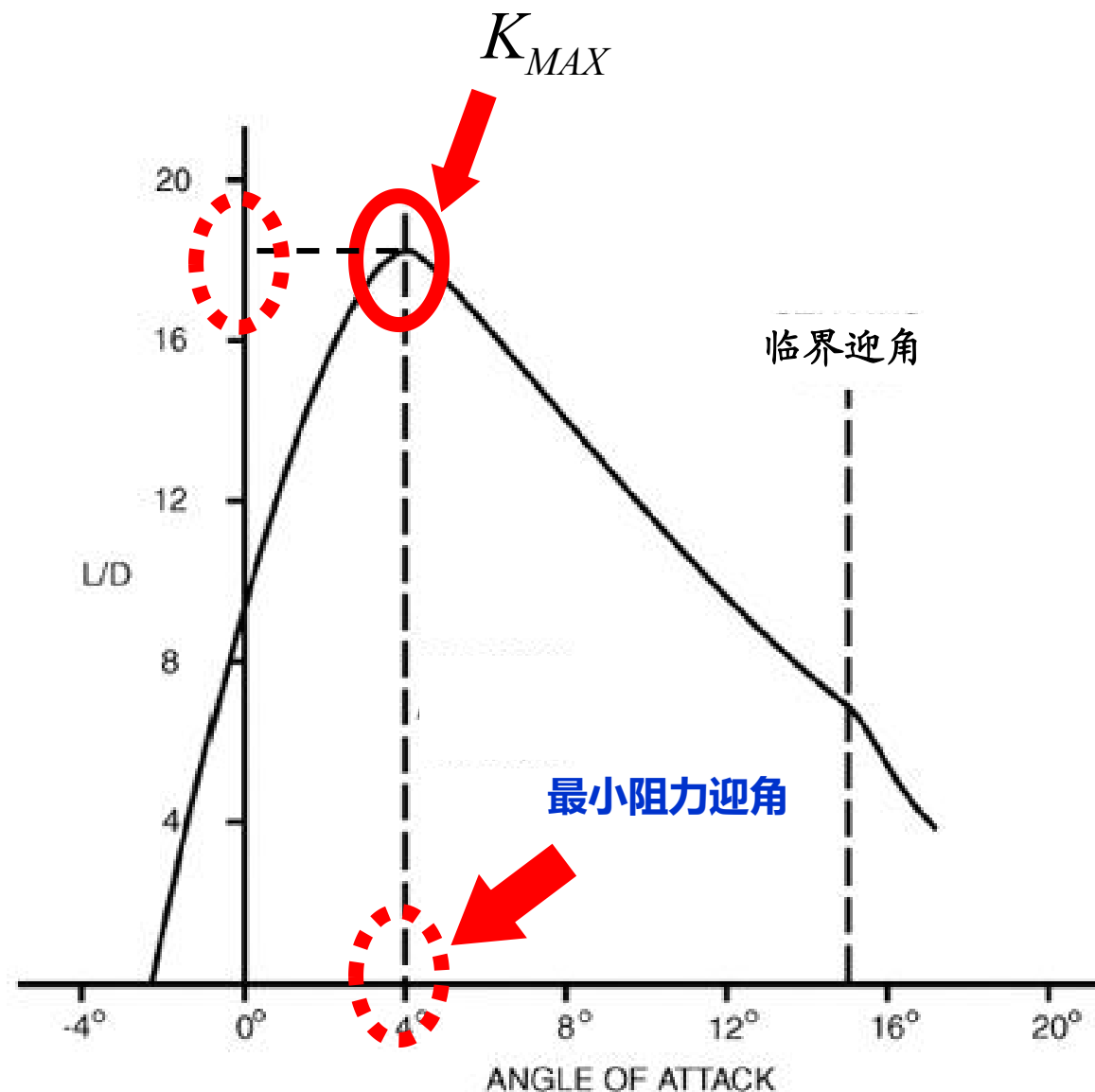
高性能滑翔机	喷气式客机	螺旋桨教练机
25~60	12~20	10~15

- 升阻比大小与飞机重量无关, 在一定范围内 (雷诺数和马赫数相似) 与空速无关, 仅与迎角相关, 在一个特定的迎角下取得最大值

升阻比曲线



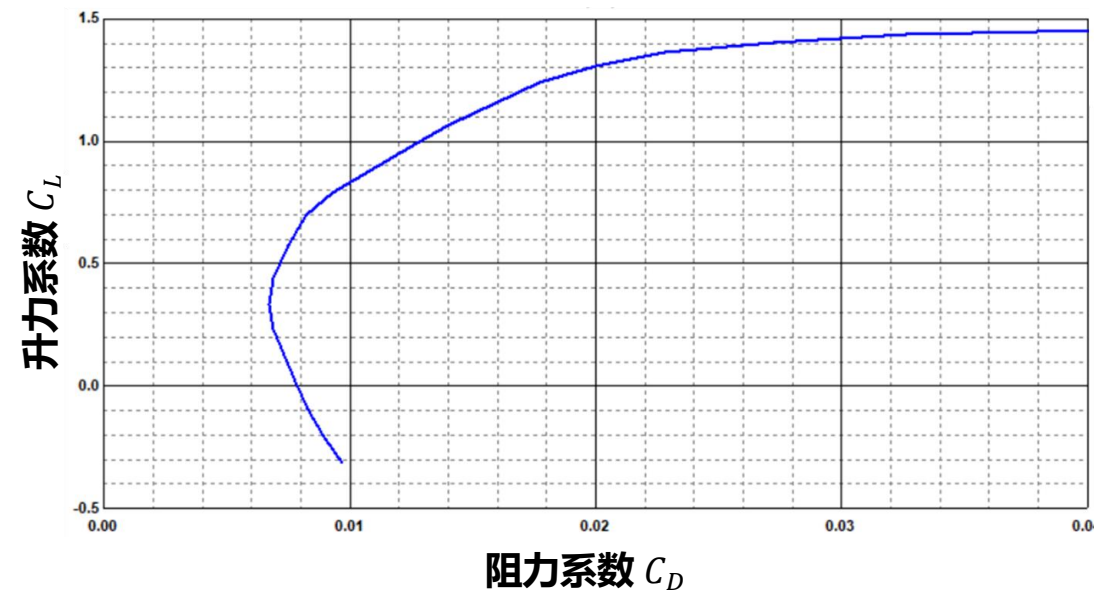
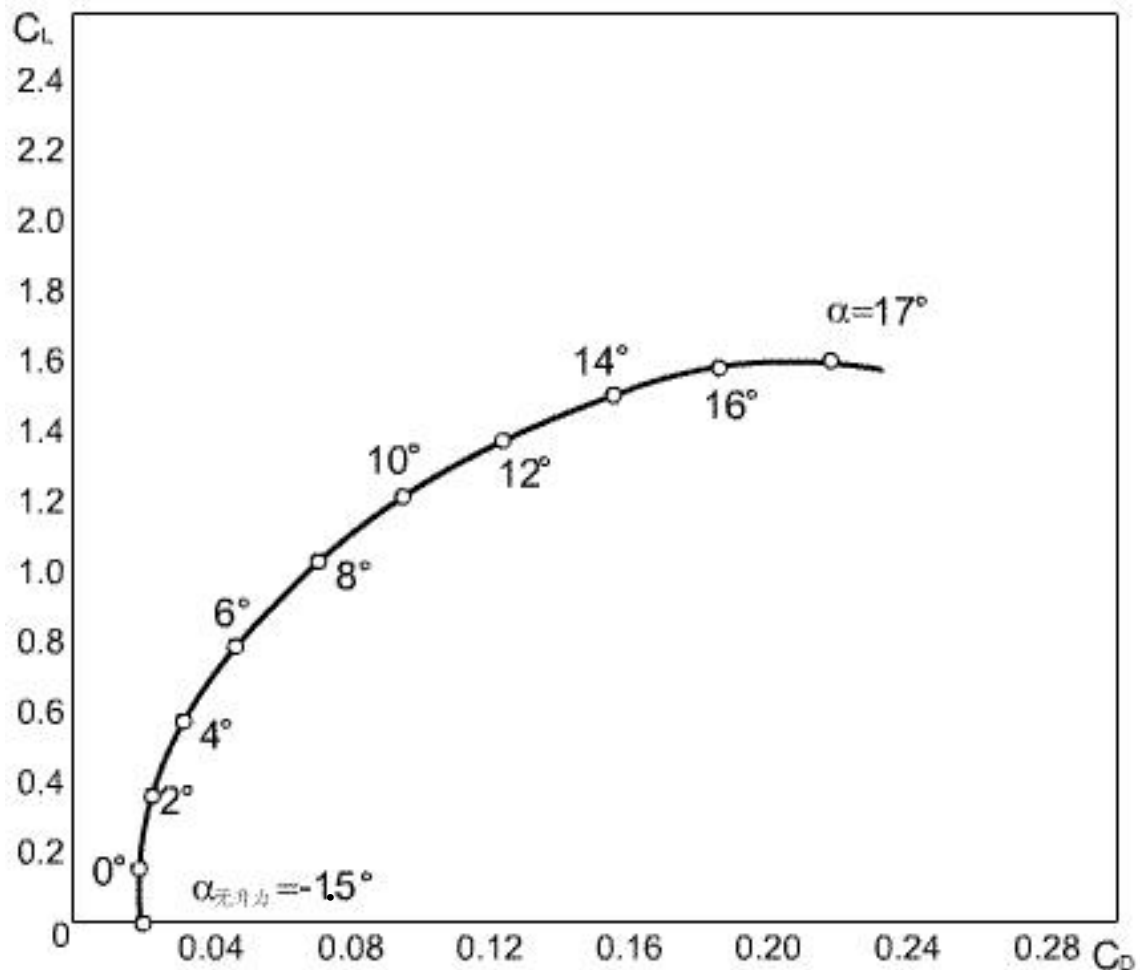
- 从零升迎角到最小阻力迎角，升力增加较快，阻力增加缓慢，因此升阻比增大。在最小阻力迎角处，升阻比最大。
- 从最小阻力迎角到临界迎角，升力增加缓慢，阻力增加较快，因此升阻比减小。
- 超过临界迎角，压差阻力急剧增大，升阻比急剧减小。



极曲线

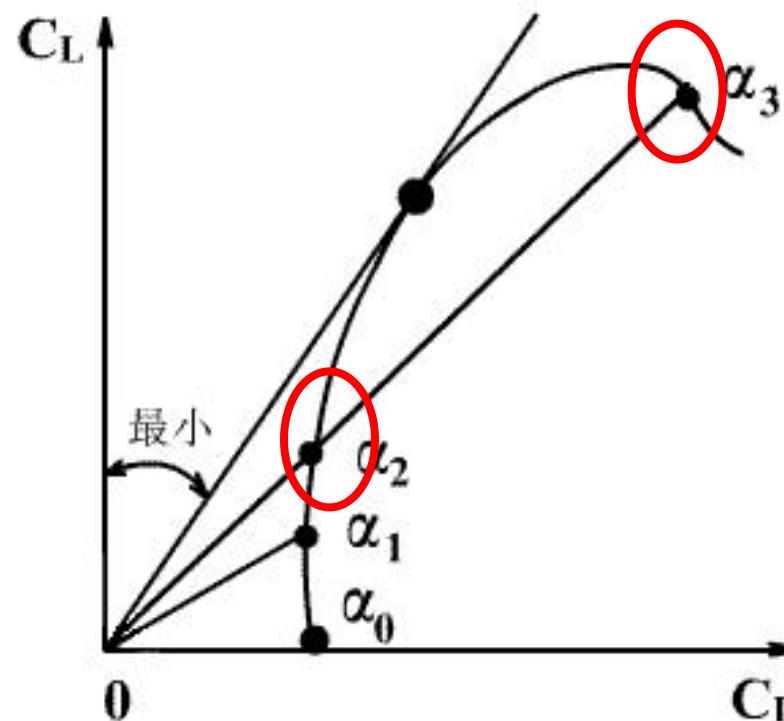
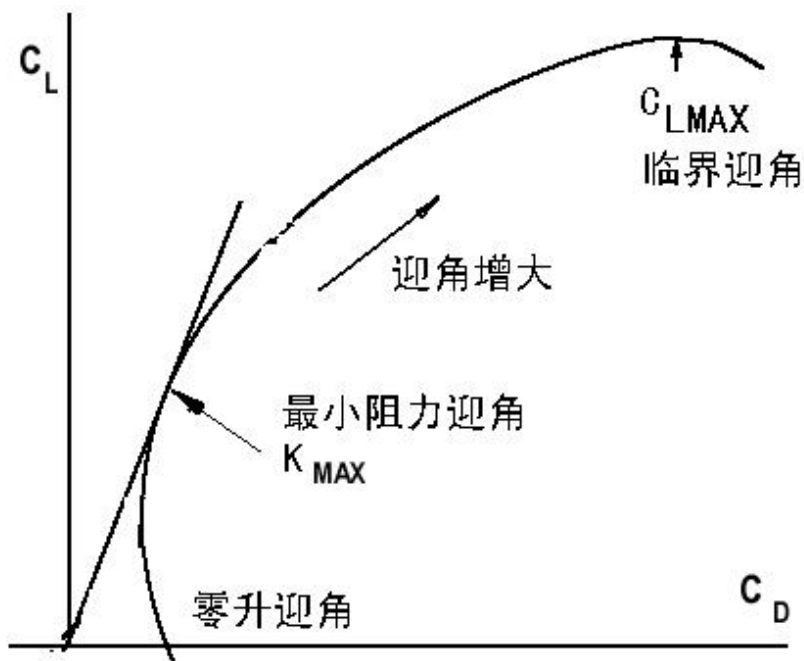


- 极曲线将飞机的升力系数、阻力系数、升阻比随迎角变化的关系综合地用一条曲线表示出来，以便于综合衡量飞机的空气动力性能





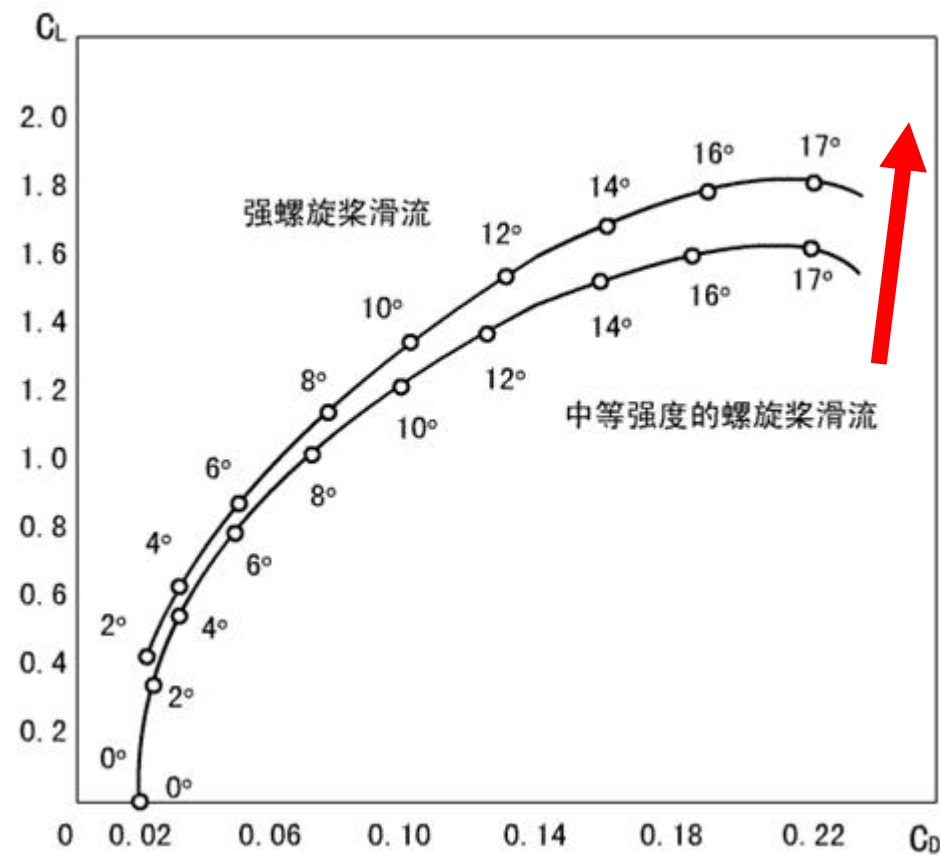
- 从坐标原点向曲线引切线，**切点对应最小阻力迎角和最大升阻比**。该迎角又称有利迎角，此时，总阻力最小，且废阻力等于诱导阻力
- 从原点所引直线与极曲线交于两点，则两点的升阻比相同，较高者的迎角较大，较高者的平飞速度较小。



极曲线的影响因素



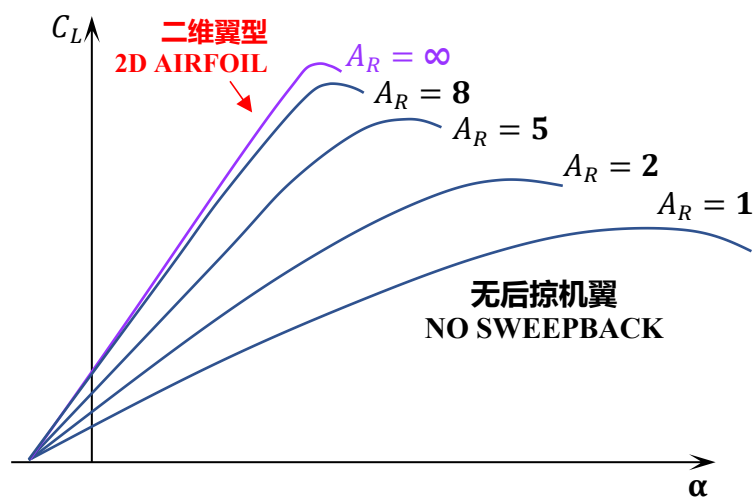
- 螺旋桨滑流使得升力系数和最大升力系数增大，最大升阻比增大，极曲线向右上偏移



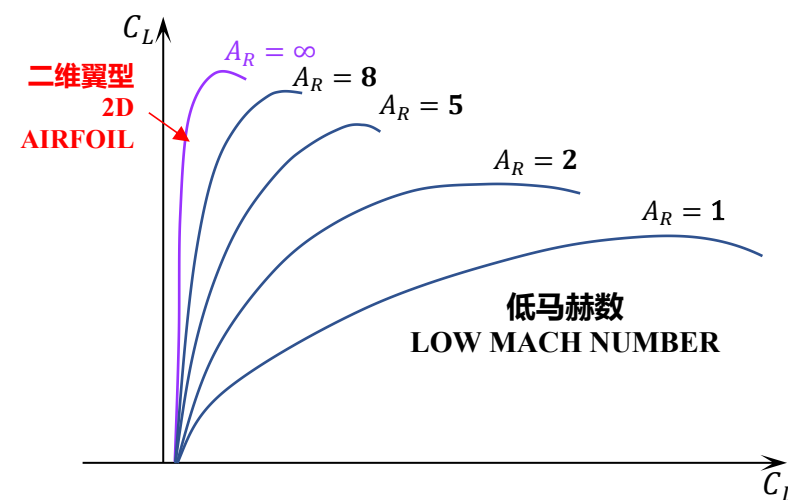
极曲线的影响因素



- 展弦比越大，低速空气动力特性越好



A_R —展弦比



低马赫数
LOW MACH NUMBER



✈ 机翼三维效应

- 翼尖涡、尾涡及尾涡面
- 下洗流场、下洗速度与诱导阻力

难点

✈ 机翼几何参数及其气动影响

- 描述机翼几何外形的参数
- 平面参数对气动的影响
- 降低诱导阻力的方式

✈ 全机低速空气动力学特性



谢 谢!

飞行原理 Principles of Flight